[INCH–POUND]
A–A–59590/8B
22 April 2010
SUPERSEDING
A–A–59590/8A
17 August 2005

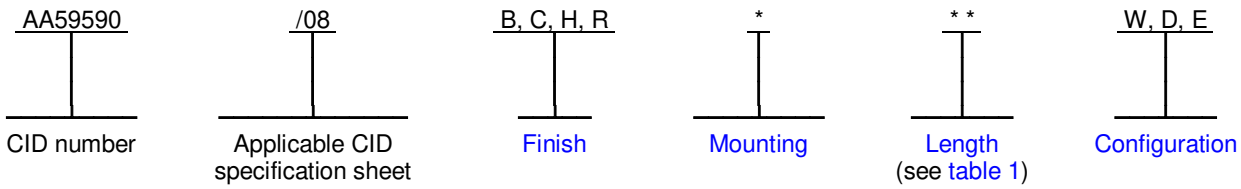
COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .220 X .220 INCH BODY SIZE, WITH VISUAL LOCK INDICATION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the issue in effect of A–A–59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).



Example: AA59590/08CM68W is the PIN for a chemical film finished, 6.8 inch (180.3 mm) long card holder with visual lock indication. The card holder also features three tapped mounting holes for use with M2.5 x.45 metric fasteners and a lockwasher and flat washer under the screw head for added resistance to loosening.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G–rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces. Card holders shall have a visual indicator to show when the assembly is in its relaxed (unlocked) state.

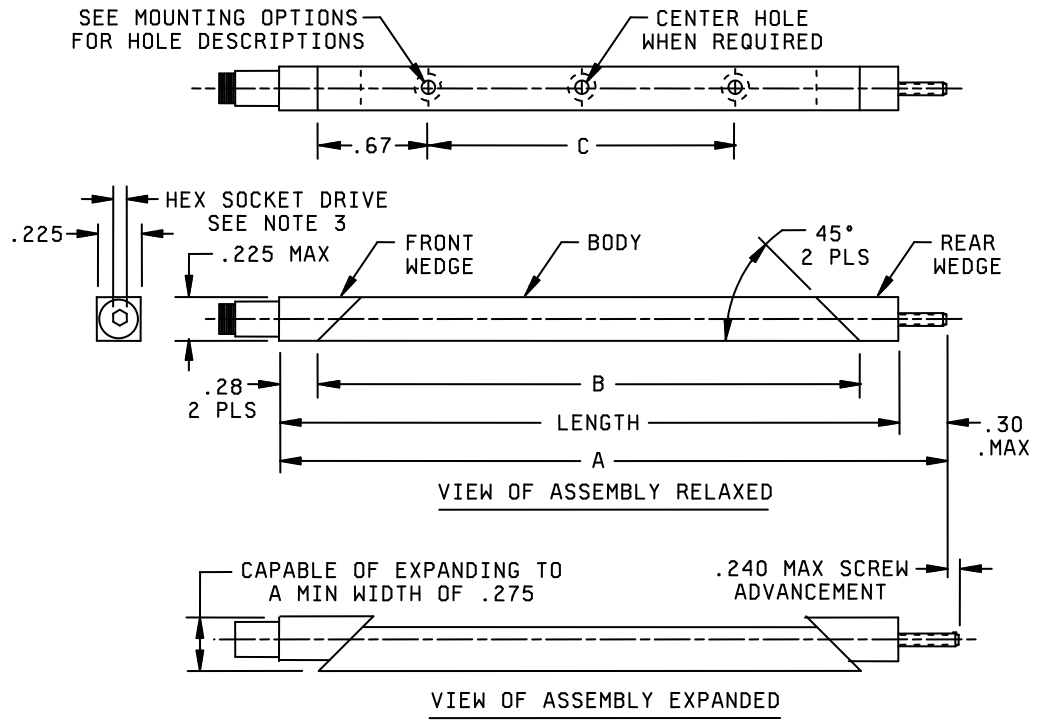
Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, 3, 4, in table 1, and A–A–59590.

Actuating screw hex drive socket. The across flats dimension for the hex drive socket shall be as follows: .094 inch (2.38 mm) for mounting options "J", "N", "T", "R", "V", and "S" and .098 inch (2.5 mm) for mounting options "G" and "M".

Material. Unless otherwise specified herein, the card holder materials shall be as specified in A–A–59590.

Visual lock indicator. The visual lock indicator, intended to show when the card holder is in its relaxed unlocked state, shall be as specified in figure 2. When the card holder is in the unlocked (relaxed) state, the end of the actuating screw shall display a red band on the side of the screw. When the actuating screw on the card holder has been tightened (assembly expanded) so that the assembly is in the locked position, this red band shall be concealed.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.225	5.72	.240	6.10	.28	7.1	.67	17.0
		.275	6.99	.30	7.62		

NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm) for mounting options "J", "N", "T", "R", "V", and "S". The across flats dimension for hex drive socket shall be a metric dimension of .098 inch (2.5 mm) across flats for mounting options "G" and "M".

FIGURE 1. Relaxed and expanded dimensions.

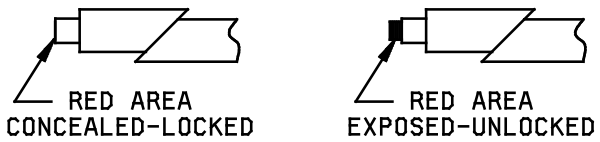


FIGURE 2. Visual lock indicator.

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Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 6 inch-pounds (0.68 N-m) for assemblies using no options or option "W" and 8 inch-pounds (0.90 N-m) for assemblies using options "E" or "D".

Cold plate slot width dimensions (when applicable). Recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .250 inch (6.35 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

Finish. The finish designator shall be as specified in A–A–59590. The finishes available for this CID specification sheet are as follows: "B" (black anodize), "C" (gold chemical film), "H" (hard black anodize), or "R" (clear chemical film).

Mounting. The mounting designators shall be as specified in A–A–59590. The mounting options available for this CID specification sheet are as follows: "J" (rivet mounting holes), "N" (no mounting holes), "T" or "R" (tapped 0-80 UNF 2B holes), "V" or "S" (tapped 2-56 UNC 2B holes), or "G" or "M" (tapped M2.5 x.45 metric holes). Card holders using mounting option "J" may be shipped unassembled. See figure 1 for mounting hole spacing requirements.

Mounting holes and locations (when required). Two mounting holes are required on card holders less than 5.30 inches (134.6 mm) in length. Three mounting holes are required on card holders 5.30 inches (134.6 mm) or greater in length. The third mounting hole shall be centered on the mounting body. See figure 1 and table I for mounting hole locations and spacing requirements.

Rivet mounting holes. The holes used for rivet mounting shall be .066 to .073 inch (1.68 to 1.85 mm) diameter through holes, countersunk 100 degrees by .140 inch (3.56 mm) diameter with an access/clearance counterbore hole of .156 inch (3.96 mm) diameter by .160 to .175 inch (4.06 to mm) deep.

Rivets. This card holder uses rivet type A as specified in A–A–59590 when rivet mounting is used.

Length, expanded, and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A–A–59590 and the available lengths for this specification sheet are listed in table I.

TABLE I. Additional assembly dimensions (see figure 1). 1/

PIN length designator	Dimension "LENGTH" ±.02 (0.5)	Dimension "A" (maximum)	Dimension "B" ±.03 (0.8)	Dimension "C" ±.005 (0.13)
18	1.8 (45.7)	2.10 (53.3)	1.24 (31.5)	.75 (19.1)
23	2.3 (58.4)	2.60 (66.0)	1.74 (44.2)	.40 (10.2)
28	2.8 (71.1)	3.10 (78.7)	2.24 (56.8)	.90 (22.9)
30	3.0 (76.2)	3.30 (83.8)	2.44 (62.0)	1.10 (27.9)
33	3.3 (83.8)	3.60 (91.4)	2.74 (69.6)	1.40 (35.6)
35	3.5 (88.9)	3.80 (96.5)	2.94 (74.7)	1.60 (40.6)
38	3.8 (96.5)	4.10 (104.1)	3.24 (82.3)	1.90 (48.3)
43	4.3 (109.2)	4.60 (116.8)	3.74 (95.0)	2.40 (61.0)
48	4.8 (121.9)	5.10 (129.5)	4.24 (107.7)	2.90 (73.7)
53	5.3 (134.6)	5.60 (142.2)	4.74 (120.4)	3.40 (86.4)
58	5.8 (147.3)	6.10 (154.9)	5.24 (133.1)	3.90 (99.1)

See footnote at end of table.

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TABLE I. Additional assembly dimensions (see figure 1) – Continued. 1/

PIN length designator	Dimension "LENGTH" ±.02 (0.5)	Dimension "A" (maximum)	Dimension "B" ±.03 (0.8)	Dimension "C" ±.005 (0.13)
63	6.3 (160.0)	6.60 (167.6)	5.74 (145.8)	4.40 (111.8)
68	6.8 (172.7)	7.10 (180.3)	6.24 (158.5)	4.90 (124.5)
73	7.3 (185.4)	7.60 (193.0)	6.74 (171.2)	5.40 (137.2)
78	7.8 (198.1)	8.10 (205.7)	7.24 (183.9)	5.90 (149.9)
83	8.3 (210.8)	8.60 (218.4)	7.74 (196.6)	6.40 (162.6)
88	8.8 (223.5)	9.10 (231.1)	8.24 (209.3)	6.90 (175.3)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

Configuration. The configuration of a card holder shall be as specified in table II. The details of a particular configuration consist of those on figures 1 and 2, and may include those on figures 3 and 4. Card holders not requiring the following option shall leave the configuration position in the PIN blank.

TABLE II. Configurations.

Configuration	Applicable figures	Hardware options
	1 and 2	No added options
W	1, 2, and 3	Lockwasher and flat washer
E	1, 2, and 4	Screw self-locking element
D	1, 2, 3, and 4	Lockwasher, flat washer and screw self-locking element

Lockwasher and flat washer (see figure 3). The use of a lockwasher and flat washer located under the screw head will provide additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer option shall include a suffix "W" in the PIN (see classification and notes).

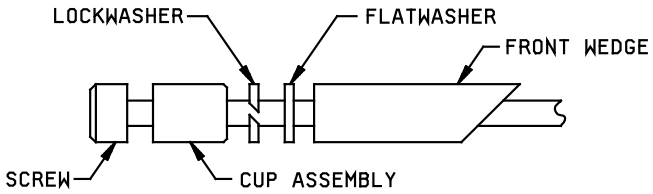


FIGURE 3. Lockwasher and flat washer details.

Screw self-locking element (see figure 4). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix "E" in the PIN (see classification and notes).

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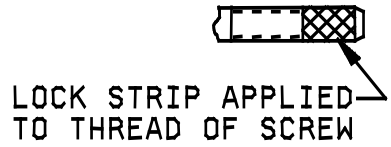


FIGURE 4. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix "D" in the PIN (see classification and notes).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Ordering data. Ordering data is as specified in A–A–59590.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	Birtcher – A Division of Pentair Electronic Packaging 7238 Trade Street San Diego, CA 92121–3410	Telephone: (858) 740–2400 Toll Free: (800) 854–7086 Facsimile: (858) 679–4555 URL: www.birtcherproducts.com
52094	Calmark – A Division of Pentair Electronic Packaging 7238 Trade Street San Diego, CA 92121–3410	Telephone: (858) 740–2400 Toll Free: (800) 854–7086 Facsimile: (858) 740–2430 Electronic mail: sales@calmark.com URL: www.calmark.com
5BG68	Card Locks Unlimited, Inc. 2310 E. Orangethorpe Avenue Anaheim, CA 92806–1231	Telephone: (714) 738–6194 Facsimile: (714) 446–0119 E-mail: sales@clumfg.com URL: www.clumfg.com

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Part number supersession data. These CID specification sheet part numbers supersede the following manufacturer's part numbers as shown in table III. The CID PINs listed in table III are only for length designator "28". See table IV for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE III. Commercial part number supersession data.

PIN designator AA59590/08	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 52094	Vendor similar designator or type part number 1/ CAGE 5BG68
BJ28	42VI–6–B	VA230–2.80H	3230VBA–2.80H
BN28	42VI–6–B–A	VA230–2.80	3230VBA–2.80
BT28	42VI–6–B–T	VA230–2.80T0	3230VBA–2.80T0
BV28	42VI–6–B–S	VA230–2.80T2	3230VBA–2.80T2
BG28	42VI–6–B–M	MVA230–2.80TM2.5	3230MVBA–2.80TM2.5
BJ28E	42VI–6–B–L	VA230–2.80HL	3230VBA–2.80HL
BN28E	42VI–6–B–A–L	VA230–2.80L	3230VBA–2.80L
BT28E	42VI–6–B–T–L	VA230–2.80T0L	3230VBA–2.80T0L
BV28E	42VI–6–B–S–L	VA230–2.80T2L	3230VBA–2.80T2L
BG28E	42VI–6–B–M–L	MVA230–2.80TM2.5L	3230MVBA–2.80TM2.5L
BJ28W	42VI–6–B–LF	WVA230–2.80H	3230VWBA–2.80H
BN28W	42VI–6–B–LF–A	WVA230–2.80	3230VWBA–2.80
BT28W	42VI–6–B–LF–T	WVA230–2.80T0	3230VWBA–2.80T0
BV28W	42VI–6–B–LF–S	WVA230–2.80T2	3230VWBA–2.80T2
BG28W	42VI–6–B–LF–M	MWVA230–2.80TM2.5	3230MVWBA–2.80TM2.5
BJ28D	42VI–6–B–LF–L	WVA230–2.80HL	3230VWBA–2.80HL
BN28D	42VI–6–B–LF–A–L	WVA230–2.80L	3230VWBA–2.80L
BT28D	42VI–6–B–LF–T–L	WVA230–2.80T0L	3230VWBA–2.80T0L
BV28D	42VI–6–B–LF–S–L	WVA230–2.80T2L	3230VWBA–2.80T2L
BG28D	42VI–6–B–LF–M–L	MWVA230–2.80TM2.5L	3230MVWBA–2.80TM2.5L
CJ28	42VI–6	V230–2.80H	3230VCG–2.80H
CN28	42VI–6–A	V230–2.80	3230VCG–2.80
CT28	42VI–6–T	V230–2.80T0	3230VCG–2.80T0
CV28	42VI–6–S	V230–2.80T2	3230VCG–2.80T2
CG28	42VI–6–M	MV230–2.80TM2.5	3230MVCG–2.80TM2.5

See footnote at end of table.

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TABLE III. Commercial part number supersession data – Continued.

PIN designator AA59590/08	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 52094	Vendor similar designator or type part number 1/ CAGE 5BG68
CJ28E	42VI–6–L	V230–2.80HL	3230VCG–2.80HL
CN28E	42VI–6–A–L	V230–2.80L	3230VCG–2.80L
CT28E	42VI–6–T–L	V230–2.80T0L	3230VCG–2.80T0L
CV28E	42VI–6–S–L	V230–2.80T2L	3230VCG–2.80T2L
CG28E	42VI–6–M–L	MV230–2.80TM2.5L	3230MVCG–2.80TM2.5L
CJ28W	42VI–6–LF	WV230–2.80H	3230VWCG–2.80H
CN28W	42VI–6–LF–A	WV230–2.80	3230VWCG–2.80
CT28W	42VI–6–LF–T	WV230–2.80T0	3230VWCG–2.80T0
CV28W	42VI–6–LF–S	WV230–2.80T2	3230VWCG–2.80T2
CG28W	42VI–6–LF–M	MWV230–2.80TM2.5	3230MVWCG–2.80TM2.5
CJ28D	42VI–6–LF–L	WV230–2.80HL	3230VWCG–2.80HL
CN28D	42VI–6–LF–A–L	WV230–2.80L	3230VWCG–2.80L
CT28D	42VI–6–LF–T–L	WV230–2.80T0L	3230VWCG–2.80T0L
CV28D	42VI–6–LF–S–L	WV230–2.80T2L	3230VWCG–2.80T2L
CG28D	42VI–6–LF–M–L	MWV230–2.80TM2.5L	3230MVWCG–2.80TM2.5L
HJ28	42VI–6–B3	VHA230–2.80H	3230VBH–2.80H
HN28	42VI–6–B3–A	VHA230–2.80	3230VBH–2.80
HT28	42VI–6–B3–T	VHA230–2.80T0	3230VBH–2.80T0
HV28	42VI–6–B3–S	VHA230–2.80T2	3230VBH–2.80T2
HG28	42VI–6–B3–M	MVHA230–2.80TM2.5	3230MVBH–2.80TM2.5
HJ28E	42VI–6–B3–L	VHA230–2.80HL	3230VBH–2.80HL
HN28E	42VI–6–B3–A–L	VHA230–2.80L	3230VBH–2.80L
HT28E	42VI–6–B3–T–L	VHA230–2.80T0L	3230VBH–2.80T0L
HV28E	42VI–6–B3–S–L	VHA230–2.80T2L	3230VBH–2.80T2L
HG28E	42VI–6–B3–M–L	MVHA230–2.80TM2.5L	3230MVBH–2.80TM2.5L
HJ28W	42VI–6–B3–LF	WVHA230–2.80H	3230WVBH–2.80H
HN28W	42VI–6–B3–LF–A	WVHA230–2.80	3230WVBH–2.80
HT28W	42VI–6–B3–LF–T	WVHA230–2.80T0	3230WVBH–2.80T0
HV28W	42VI–6–B3–LF–S	WVHA230–2.80T2	3230WVBH–2.80T2
HG28W	42VI–6–B3–LF–M	MWVHA230–2.80TM2.5	3230MVWBH–2.80TM2.5

See footnote at end of table.

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TABLE III. Commercial part number supersession data – Continued.

PIN designator AA59590/08	Vendor similar designator or type part number 1/ CAGE 18915	Vendor similar designator or type part number 1/ CAGE 52094	Vendor similar designator or type part number 1/ CAGE 5BG68
HJ28D	42VI–6–B3–LF–L	WVHA230–2.80HL	3230VWBH–2.80HL
HN28D	42VI–6–B3–LF–AL	WVHA230–2.80L	3230VWBH–2.80L
HT28D	42VI–6–B3–LF–TL	WVHA230–2.80T0L	3230VWBH–2.80T0L
HV28D	42VI–6–B3–LF–SL	WVHA230–2.80T2L	3230VWBH–2.80T2L
HG28D	42VI–6–B3–LF–ML	MWVHA230–2.80TM2.5L	3230MVWBH–2.80TM2.5L
RJ28	42VI–6–CC	VR230–2.80H	3230VCC–2.80H
RN28	42VI–6–CC–A	VR230–2.80	3230VCC–2.80
RT28	42VI–6–CC–T	VR230–2.80T0	3230VCC–2.80T0
RV28	42VI–6–CC–S	VR230–2.80T2	3230VCC–2.80T2
RG28	42VI–6–CC–M	MVR230–2.80TM2.5	3230MVCC–2.80TM2.5
RJ28E	42VI–6–CC–L	WVR230–2.80H	3230VCC–2.80HL
RN28E	42VI–6–CC–A–L	WVR230–2.80	3230VCC–2.80L
RT28E	42VI–6–CC–T–L	WVR230–2.80T0	3230VCC–2.80T0L
RV28E	42VI–6–CC–S–L	WVR230–2.80T2	3230VCC–2.80T2L
RG28E	42VI–6–CC–M–L	MWVR230–2.80TM2.5	3230MVCC–2.80TM2.5L
RJ28W	42VI–6–CC–LF	WVR230–2.80HL	3230WVCC–2.80H
RN28W	42VI–6–CC–LF–A	WVR230–2.80L	3230WVCC–2.80
RT28W	42VI–6–CC–LF–T	WVR230–2.80T0L	3230WVCC–2.80T0
RV28W	42VI–6–CC–LF–S	WVR230–2.80T2L	3230WVCC–2.80T2
RG28W	42VI–6–CC–LF–M	MWVR230–2.80TM2.5L	3230MVWCC–2.80TM2.5
RJ28D	42VI–6–CC–LF–L	WVR230–2.80HL	3230VWCC–2.80HL
RN28D	42VI–6–CC–LF–AL	WVR230–2.80L	3230VWCC–2.80L
RT28D	42VI–6–CC–LF–TL	WVR230–2.80T0L	3230VWCC–2.80T0L
RV28D	42VI–6–CC–LF–SL	WVR230–2.80T2L	3230VWCC–2.80T2L
RG28D	42VI–6–CC–LF–ML	MWVR230–2.80TM2.5L	3230MVWCC–2.80TM2.5L

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

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TABLE IV. Example of PIN with available length designators.

PIN designator AA59590/08	Vendor similar designator or type part number 1/ 2/ CAGE 18915	Vendor similar designator or type part number 1/ 2/ CAGE 52094	Vendor similar designator or type part number 1/ 2/ CAGE 5BG68
CG18W	42VI–4–LF–M		3230WCG–1.80HL
CG23W	42VI–5–LF–M	MV230–2.30TM2.5	3230WCG–2.30HL
CG28W	42VI–6–LF–M	MV230–2.80TM2.5	3230WCG–2.80HL
CG30W	3/	MV230–3.00TM2.5	3230WCG–3.00HL
CG33W	42VI–7–LF–M	MV230–3.30TM2.5	3230WCG–3.30HL
CG35W	3/	MV230–3.50TM2.5	3230WCG–3.50HL
CG38W	42VI–8–LF–M	MV230–3.80TM2.5	3230WCG–3.80HL
CG43W	42VI–9–LF–M	MV230–4.30TM2.5	3230WCG–4.30HL
CG48W	42VI–10–LF–M	MV230–4.80TM2.5	3230WCG–4.80HL
CM53W	42VI–11–LF–M	MV230–5.30ETM2.5	3230WCG–5.30EHL
CM58W	42VI–12–LF–M	MV230–5.80ETM2.5	3230WCG–5.80EHL
CM63W	42VI–13–LF–M	MV230–6.30ETM2.5	3230WCG–6.30EH
CM68W	42VI–14–LF–M	MV230–6.80ETM2.5	3230WCG–6.80EHL
CM73W	42VI–15–LF–M	MV230–7.30ETM2.5	3230WCG–7.30EHL
CM78W	42VI–16–LF–M	MV230–7.80ETM2.5	3230WCG–7.80EHL
CM83W	42VI–17–LF–M	MV230–8.30ETM2.5	3230WCG–8.30EHL
CM88W	42VI–18–LF–M	MV230–8.80ETM2.5	3230WCG–8.80EHL

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.
3/ Consult manufacturer for part number.

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MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Review Activities:
Army – MI
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FAS

Preparing Activity:
DLA – CC

Project 5998–2009–056

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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